S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Session-2025-26 (Even)

TAE-I Project Based Learning

Topic Name: Bayesian Network for Disease Prediction

Submitted By

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AI Technique(s) (2)	Problem Representation (4)	Tools Explored (2)	Outcome (2)	Total (10)
Sign of Student:			Sign of course In-charge :	

1. Industry Problem Context

In today's fast-paced world, access to timely and accurate healthcare remains a major challenge. Many individuals ignore early symptoms due to lack of awareness, time constraints, or limited access to medical professionals. This often leads to:

- Late diagnosis of diseases
- Increased healthcare costs
- Overcrowded hospitals
- Higher risk of complications

In the IT and healthcare industries, there is a growing need for intelligent systems that can assist users in identifying diseases at an early stage based on symptoms.

Where this problem exists:

- Telemedicine platforms
- Online health consultation systems
- Hospital triage systems
- Mobile health monitoring applications
- AI-based healthcare assistants

The main goal is to **analyze symptoms and predict possible diseases quickly, efficiently, and accurately**, helping users take early action.

2. Feasibility in IT Systems

The implementation of a disease prediction system is highly feasible using modern IT infrastructure due to the availability of:

- Large medical datasets
- Powerful machine learning algorithms
- Cloud computing resources
- Web-based deployment technologies

System Modeling Feasibility

The problem can be modeled as a **classification problem**:

- **Input:** Symptoms (e.g., fever, headache, cough)
- **Processing:** Machine Learning Model
- **Output:** Predicted disease

Data Representation

- Symptoms → Features
- Diseases → Labels
- Dataset → Structured table

Why it is scalable:

- Can handle large datasets
- Can be deployed on cloud platforms
- Can support multiple users simultaneously
- Easy integration with APIs and databases

Thus, such systems are not only feasible but already widely used in industry.

3. AI Techniques Used in Industry

The Disease Predictor system is based on **Artificial Intelligence and Machine Learning techniques**, specifically **supervised learning**.

Techniques Used

1. Classification Algorithms

These algorithms classify input symptoms into a specific disease category.

Common algorithms:

- Decision Tree
- Naive Bayes
- Random Forest

2. Supervised Learning

- Model is trained using labeled data
- Learns mapping between symptoms and diseases

3. Pattern Recognition

- Identifies patterns in symptoms
- Matches them with known disease patterns

Working Principle

- Input symptoms are converted into numerical format
- Model processes input using trained data
- Predicts most probable disease

Advantages

- Fast prediction
- Good accuracy
- Easy to train and deploy

These techniques are similar to those used in advanced healthcare AI systems like IBM Watson Health.

4. Representation / Modeling in Industry Systems

System Representation

The system is modeled as a **Machine Learning Pipeline**:

Components:

1. Input Layer (Symptoms)
2. Processing Layer (ML Model)
3. Output Layer (Predicted Disease)

Data Modeling

- Each symptom is represented as:
  - Binary value (0 or 1) OR
  - Weighted value
- Example:

```
Fever = 1
Cough = 1
Headache = 0
```

Workflow

1. User inputs symptoms
2. Data is preprocessed
3. Model evaluates input
4. Disease is predicted
5. Result displayed

System Characteristics

- Dynamic and updatable
- Can incorporate new diseases
- Can improve with more data

5. Tools & Technologies Used

Programming Languages

- Python (core backend logic)

Frontend

- HTML
- CSS
- JavaScript

Backend Framework

- Flask (lightweight web framework)

Libraries Used

- Pandas → Data handling
- NumPy → Numerical computation
- Scikit-learn → Machine learning

Development Platform

- GitHub for hosting
- Local server for testing

Industry Comparison

- Python is widely used in AI development
- Flask/Django used for web deployment
- Scikit-learn used for ML prototyping

6. Industry-Level Outcome

The developed system achieves:

Key Outcomes

- Quick disease prediction
- Reduced manual diagnosis effort
- Increased accessibility to healthcare

Performance Metrics

- Accuracy of prediction
- Response time
- Efficiency of model

Benefits

- Saves time
- Reduces workload on doctors
- Provides initial guidance to users

7. Real-World Industry Examples / Use

Many companies and platforms use similar systems:

- Practo → Online doctor consultation
- Ada Health → AI-based diagnosis
- IBM Watson Health → Advanced AI diagnosis

Applications

- Telemedicine
- Self-diagnosis tools
- Hospital assistance systems
- Health monitoring apps

8. Implementation & Result

Project Description

The **Disease Predictor Web Application** is a Machine Learning-based system that predicts diseases based on symptoms entered by the user.

System Architecture

```
User Interface → Backend (Flask) → ML Model → Output
```

Step-by-Step Implementation

Step 1: Dataset Preparation

- Collected medical dataset
- Cleaned and structured data
- Converted symptoms into features

Step 2: Model Training

- Used classification algorithm
- Trained model on dataset
- Evaluated accuracy

Step 3: Backend Development

- Flask used to handle requests
- Connected ML model to server

Step 4: Frontend Development

- Created user-friendly interface
- Input form for symptoms
- Display prediction result

Step 5: Integration

- Connected frontend with backend
- Tested complete workflow

Features of the System

- Easy-to-use interface
- Fast prediction
- Accurate results
- Lightweight design
- Accessible from browser

Result

- The system successfully predicts diseases based on input symptoms
- Provides instant output
- Works efficiently with minimal delay

Limitations

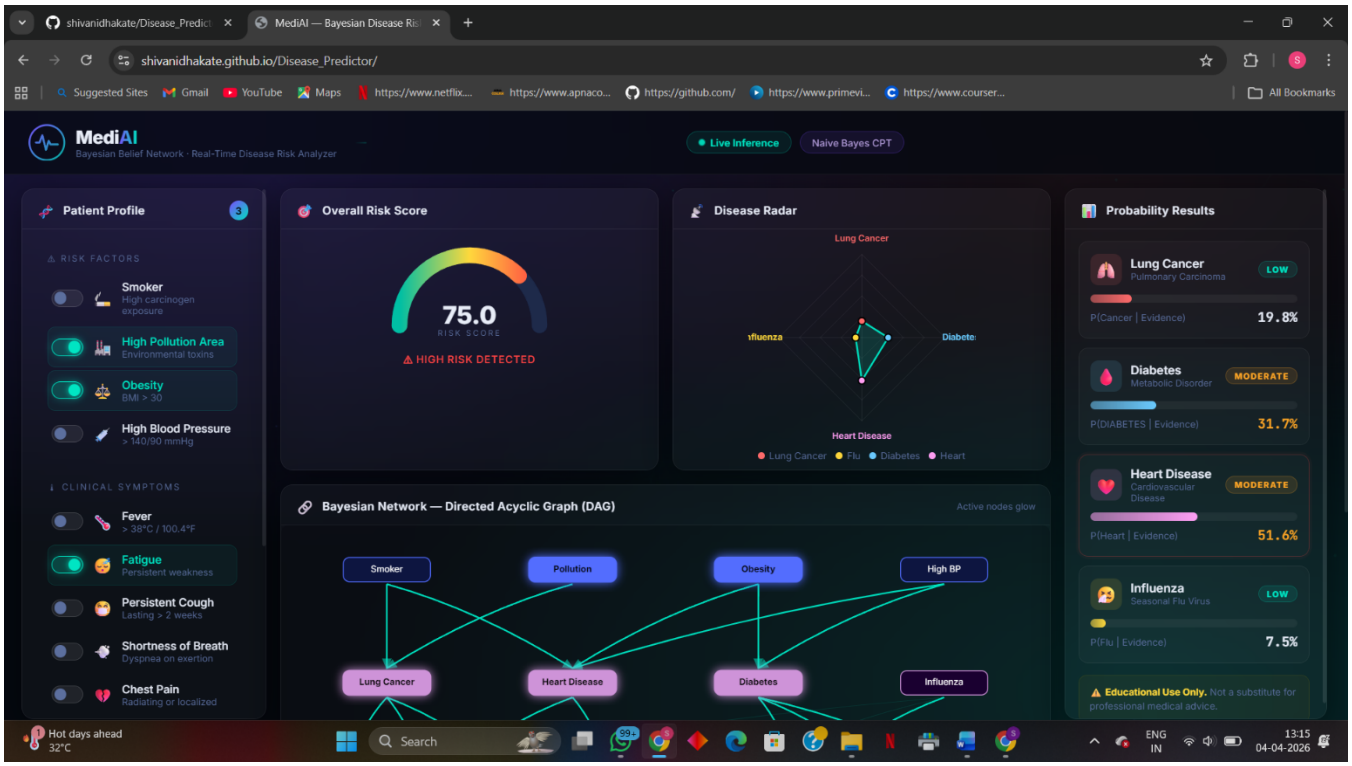
- Depends on dataset quality
- Cannot replace professional medical advice
- Limited to predefined diseases

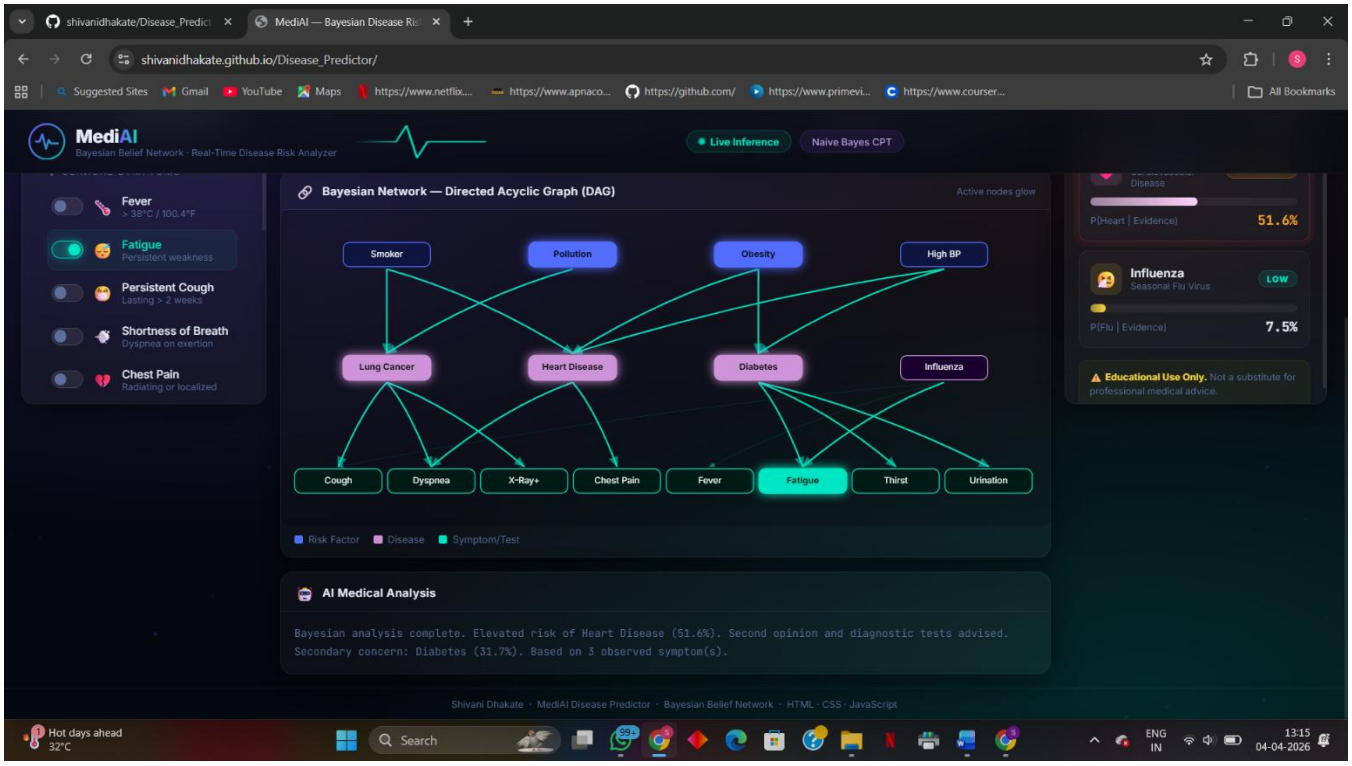
Future Enhancements

- Add more diseases
- Improve accuracy with deep learning
- Add chatbot support
- Integrate with real-time medical APIs

Github link : https://github.com/shivanidhakate/Disease_Predictor

Host link: https://shivanidhakate.github.io/Disease_Predictor/





9. References

- Scikit-learn Official Documentation
- Python Official Documentation
- Flask Documentation
- Kaggle Medical Datasets
- Research papers on AI in healthcare
- IBM Watson Health resources

Conclusion

The Disease Predictor System demonstrates the effective use of Artificial Intelligence in healthcare. It provides a simple, fast, and efficient way to predict diseases based on symptoms. While it cannot replace doctors, it serves as a valuable tool for early diagnosis and awareness.

This project highlights how AI can be integrated into real-world applications to solve critical problems and improve human life.